Active Maintenance Report: coppock_2014

Coppock (2014, Journal of Experimental Political Science): Information Spillovers, and its 2016 corrigendum

2026-08-01

Table of contents

1 Summary **3**
- 1.1 Does the deposited archive run? . . . 3
- 1.2 Does the maintained rewrite reproduce the paper? . . . 3

2 Paper overview **4**

3 Original archive reproducibility **5**
- 3.1 The directory that is not in the deposit . . . 5
- 3.2 The R 3.6 sampler . . . 6
- 3.3 Packages . . . 6

4 The corrigendum **6**

5 Errata **8**
- 5.1 The point estimates of Figure 3 . . . 8
- 5.2 The seed that is not set . . . 9
- 5.3 The simulation that never permutes . . . 9

6 Number-by-number comparison **10**

7 Maintained rewrite **11**
- 7.1 Architecture . . . 11
- 7.2 Deprecated patterns replaced . . . 12
- 7.3 Summary tables . . . 12
- 7.4 In-text quantities . . . 13

8 Figure verification **13**

9 Maintained rewrite verification 20

10 R environment 22

This repository holds the actively maintained replication code for Coppock (2014) and for the corrigendum that corrects it, together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/xps.2014.9
Corrigendum	10.1017/xps.2016.8
Replication archive	10.7910/DVN/28664
Reanalyzed experiment	Butler and Nickerson (2011), 10.1561/100.00011019

The corrigendum is a separately published article with its own DOI, its own year and its own pagination, so it is cited here in its own right rather than as an attachment to the 2014 paper. It matters for what follows: the deposited archive was never updated after it appeared, so the code at the DOI above still produces the results the corrigendum retracts.

The data are not redistributed here. The deposit is 3.0 MB across 30 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published figure or table, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_2014.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 30 files, and produces every figure and table into `maintained/output/`. Required packages: tidyverse, sf, spdep, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about five minutes, most of it in the four-parameter appendix model. A successful run overwrites

`maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Not as deposited, and the reason is structural rather than a matter of package rot. Every one of the seven figure scripts opens with `load("Replication Archive/CoppockJEPS.rdata")` and writes into `Replication Archive/figures/` and `Replication Archive/modeloutputs/`. No such directory is in the deposit: all 30 files sit flat at the top level. The archive's own guide, in section 4 of the online appendix, refers to a `CoppockJEPS_shapefile` directory that "must stay intact", so the folder structure the code expects existed on the author's machine and did not survive upload. A reader has to build three directories and copy a file into one of them before any script runs. That is a five-minute repair, and it is invisible until you attempt it.

Once the directories exist, the seven figure scripts run without further intervention. The data preparation script that builds their input does not: it calls `maptools`, which left CRAN in 2023. One of the seven, `CoppockJEPS_appendixfigure4.R`, is the only figure script in the archive that never calls `set.seed`, so Appendix Figure 4 gives a different answer on every run and cannot be reproduced by anyone, its author included.

The scripts then reproduce the article. 42 of the 47 article claims an archive script can be checked against come back exactly as printed on a current R, and every one of the remaining 5 is a confidence region endpoint one grid step away, moved there by the change R 3.6 made to `sample()`. Restoring the old sampler recovers all 5, and costs one: Appendix Figure 4's direct effect matches on a current R and is a step off under the old one, which is what a figure whose script sets no seed does.

What the archive does not contain is any trace of the correction. The deposited `CoppockJEPS.RData` still holds the corrupted similarity matrix described below, so the figures a reader produces today are the ones the corrigendum retracts.

1.2 Does the maintained rewrite reproduce the paper?

It reproduces the 2014 article and it does not reproduce the 2016 corrigendum, and the second half of that sentence is the finding.

The rewrite reproduces the article's p-value maps exactly. Where a map can be compared against the deposited scripts cell by cell it is bit-identical: Figures 1a, 1b, 2, 3, A2 and A3 all

come back with the same p-value matrix the archive produces, to the last bit. The published values it does not return are of three kinds, and each has a cause the ground truth records: five confidence region endpoints that R 3.6 moved one grid step, three Figure 3 point estimates that the archive reads off the wrong element of an index matrix, and all of Appendix Figure 4, whose script sets no seed.

The corrigendum is a different matter. It reports that the ideological similarity matrix used in Figures 2 and 3 wrongly implied that all legislators were equally close, gives corrected estimates of -0.250 and -0.150 for the direct and indirect effects, and concludes that the evidence for spillovers is insufficient. The deposit does contain the corrupted matrix the corrigendum describes. Rebuilding it from Equation (1) of the paper and rerunning does widen both confidence regions across zero, which is the corrigendum's conclusion. It does not return the corrigendum's numbers: the corrected point estimates come back at the same place the uncorrected ones did, with confidence regions narrower than the published corrected ones. **The corrigendum's arithmetic cannot be reproduced from the deposited archive**, because the archive was never updated to hold the code that produced it.

The rewrite also turns up three coding errors the corrigendum does not mention: an index that reads Figure 3's point estimates off the wrong element of a matrix, a missing seed in Appendix Figure 4, and a simulation loop in Appendix Figure 5 that never permutes. See "Errata".

2 Paper overview

Citation: Coppock, A. (2014). "Information spillovers: Another look at experimental estimates of legislator responsiveness." *Journal of Experimental Political Science*, 1(2), 159-169. DOI: 10.1017/xps.2014.9

Corrigendum: Coppock, A. (2016). *Journal of Experimental Political Science*, 3(2), 206-208. DOI: 10.1017/xps.2016.8

Summary: Butler and Nickerson (2011) mailed district-level public opinion data to 35 of the 70 members of the New Mexico House before a 2008 vote on a tax rebate bill, in a matched-pair randomization, and estimated the effect of that information on roll call votes. The reanalysis asks what happens when the non-interference assumption is dropped, on the grounds that legislatures are information-sharing networks. Legislators are assumed to be exposed to treatment through their ideologically similar colleagues, with similarity measured from W-NOMINATE ideal points estimated on the preceding session's roll calls. Inference follows Bowers, Fredrickson and Panagopoulos (2013): each pair of hypothesized direct and indirect effects implies a uniformity trial, the hypothesis is re-imposed under draws from the 10,000 admissible randomizations, and the p-value is the share of draws whose sum of squared residuals the observed one exceeds. The article concluded that the total causal effect of treatment was about twice the difference in means. The corrigendum retracts the spillover part of that conclusion.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
CoppockJEPS_10000Randomizations.R	Runs; regenerates a matrix the deposit already carries, under a seed R has since changed	None needed: Z_block is inside the deposited CoppockJEPS.RData
CoppockJEPS_datapreparation.R	Fails: maptools is off CRAN, and it writes CoppockJEPS.rdata over the deposited CoppockJEPS.RData	sf::st_read plus spdep::poly2nb rebuilds the one matrix maptools was used for
CoppockJEPS_figure1acode.R	Fails on a missing directory; runs once it exists	Create Replication Archive/figures and .../modeloutputs; copy CoppockJEPS.RData in
CoppockJEPS_figure1bcode.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_figure2code.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_figure3code.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_appendixfigure2.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_appendixfigure3.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_appendixfigure4.R	Fails on a missing directory; runs once it exists, and sets no seed	As above; set.seed(343), as every other script does
CoppockJEPS_appendixfigure5.R	Fails on a missing directory; runs once it exists, and its simulation never permutes	As above; draw a fresh assignment per simulation
CoppockJEPS_appendixtable.R	Fails: loads seven files by bare name and writes to a hardcoded /Users/Alex path	Run from the modeloutputs directory; redirect the output

3.1 The directory that is not in the deposit

The scripts and the deposit disagree about where files live, in both directions. Every figure script reads `Replication Archive/CoppockJEPS.rdata` and writes to `Replication Archive/figures/`; the data preparation script reads `CoppockJEPSshapefile/hd_court_ordered.shp`. Neither path exists at Dataverse, where the shapefile's three components sit at the top level alongside everything else. The published appendix documents the shapefile directory as a directory, so this is an upload that flattened a tree rather than an author who never had one.

Two smaller path defects come with it, and both are invisible on macOS.

`CoppockJEPS_10000Randomizations.R` writes `CoppockJEPS_10000randomizations.rdata`, with a lower-case r, and `CoppockJEPS_datapreparation.R` loads `CoppockJEPS_10000Randomizations.rdata`, with an upper-case one. A case-insensitive filesystem resolves that; Linux does not.

More seriously, `CoppockJEPS_datapreparation.R` ends by writing `CoppockJEPS.rdata` into the working directory, and the deposited file is `CoppockJEPS.RData`. Running the data preparation script inside a copy of the archive on macOS therefore silently overwrites a deposited file with a regenerated one. That is why everything in this report was run from a scratch copy, and why `download_original.R` reports anything in `original/` that the manifest does not list.

3.2 The R 3.6 sampler

Figures 2, 3, A2, A3 and A4 draw their simulated assignments with `Z_block[, sample(1:10000, 1)]` after a single `set.seed(343)`. R 3.6.0 replaced the rounding method `sample()` used to turn uniform draws into integers with a rejection method, so the same seed now walks a different path through the permutation matrix. Figures 1a and 1b are immune, because they take the first hundred columns in order rather than sampling.

Under the modern sampler 5 of the article's claims come back one grid step away from where they were printed, and `RNGkind(sample.kind = "Rounding")` recovers every one. The ground truth records the pre-3.6 value in its notes column for each of the 7 quantities that take different values under the two samplers. Nothing about the analysis is wrong in either case: the p-value surface is estimated from 500 draws per cell, its maximum is a plateau rather than a point, and a confidence region read off a 0.025 grid moves by a step when the draws change. The rewrite keeps the modern sampler and this report records the drift rather than suppressing it.

3.3 Packages

Two package findings, and the March 2026 pass got both of them backwards.

`maptools` is genuinely gone: it was archived on CRAN in October 2023 along with `rgdal` and `rgeos`. It was used for exactly one line, `readShapeSpatial` on the district shapefile, and only for the geographic contiguity placebo in Appendix Figure 4. It did not build the ideological similarity matrices, which come from `wnominate`. `sf::st_read` followed by `spdep::poly2nb` reads the same deposited shapefile and returns an adjacency matrix identical to the deposited one, entry for entry, so nothing in the paper depends on a package that no longer exists.

`wnominate` is the opposite case. It is still on CRAN, at version 1.5, and installs without incident, so the ideal points can be re-estimated. They do not come back the same. Re-estimating from the deposited roll call data correlates 0.98 with the deposited scores, with a median absolute difference of 0.12 and a maximum of 0.48 on a scale running from -1 to 1. The rewrite therefore reads the deposited ideal points rather than re-estimating them, which keeps the similarity matrices identical to the ones the paper used; regenerating them would be a different analysis, not a reproduction of this one.

4 The corrigendum

The corrigendum is the substance of the case, so it is worth being precise about which object is wrong.

`CoppockJEPS.RData` stores four similarity matrices. Three of them, the rank, squared-distance and geographic ones, rebuild from the deposited ideal points and shapefile exactly. The fourth, `similarity.matrix.dw`, is 1 in every off-diagonal cell and 0 on the diagonal. It asserts

that every pair of legislators in the New Mexico House is maximally and identically close in ideological space, which is precisely what the corrigendum says went wrong.

The corruption is confined to that one object. The net exposure variable the published regressions use, `s.difs.dw`, was computed from a correct similarity matrix: rebuilding it from Equation (1) reproduces the deposited values to fourteen decimal places. The published analysis therefore regressed the observed outcome on a correctly built exposure measure while simulating the null distribution with the all-ones matrix. Under that matrix every unit's raw exposure is the number of treated colleagues, which is the same for everyone, so the simulated exposure carries no variation attributable to the random assignment at all. The reference distribution is far too narrow, and the confidence regions that come out of it are far too tight. That is the mechanism behind the corrigendum.

Table 2: Figure 2 as the corrigendum states it, as the deposited scripts produce it, and as the maintained rewrite produces it. A blank archive column means the deposit contains no corrected analysis to run.

Version	Quantity	Corrigendum	Archive	Rewrite
Uncorrected	direct effect	-0.3	-0.3	-0.3
Uncorrected	direct 95% CI lower	-0.575	-0.5	-0.5
Uncorrected	direct 95% CI upper	0	-0.075	-0.075
Uncorrected	indirect effect	-0.225	-0.225	-0.225
Uncorrected	indirect 95% CI lower	-0.325	-0.325	-0.325
Uncorrected	indirect 95% CI upper	-0.025	-0.1	-0.1
Corrected	direct effect	-0.25		-0.2875
Corrected	direct 95% CI lower	-0.625		-0.5
Corrected	direct 95% CI upper	0.175		0.075
Corrected	indirect effect	-0.15		-0.225
Corrected	indirect 95% CI lower	-0.325		-0.3
Corrected	indirect 95% CI upper	0.2		0.15

Three things in that table are worth separating.

The corrigendum's account of the uncorrected result does not match the uncorrected result. It reports the original 95% confidence intervals as running from -0.325 to -0.025 for the indirect effect and from -0.575 to 0.000 for the direct effect. Appendix Table 1 of the 2014 article gives [-0.3, -0.1] and [-0.5, -0.075], and those are what the deposited scripts produce today. The point estimates agree; the interval endpoints are three grid steps apart on the direct effect. The most likely explanation is that the corrigendum re-ran the uncorrected model rather than quoting the article, and got a differently seeded surface, which is the same instability documented above. It is nonetheless a discrepancy inside the published record: two articles state different confidence intervals for the same estimate.

The correction reproduces qualitatively and not numerically. Rebuilding the ideological similarity matrix and rerunning is a one-line change to the deposited code, and it does what the corrigendum says: the direct interval widens to include zero, and the indirect interval

widens to include zero. The 95% regions come back at [-0.5, 0.075] and [-0.3, 0.15] rather than the published [-0.625, 0.175] and [-0.325, 0.2], and the corrected point estimates stay at the uncorrected values rather than moving to -0.25 and -0.15.

That gap is not Monte Carlo noise. Running the corrected model at five different seeds moves the direct point estimate between -0.3 and -0.2875 and leaves the intervals within one grid step of each other; ten times the number of simulations lands in the same place, as do a hundredth and a tenth of them, a grid widened to the -0.75 to 0.25 range the corrigendum's axis labels suggest, and leaving the similarity matrix's diagonal unzeroed. The corrigendum's numbers are outside the range every one of those produces. Something else about the corrected run differed, and the deposit does not record what: the archive was released in 2015 and never revised, so the code behind the corrigendum's Figure 1 exists nowhere public. **A corrigendum that corrects a published analysis is itself a published analysis, and this one has no archive.**

5 Errata

Three coding errors, none of them the one the corrigendum reports.

5.1 The point estimates of Figure 3

The paper takes the Hodges-Lehmann point estimate to be the hypothesis pair with the highest p-value. Where several cells tie for the maximum, `CoppockJEPS_figure2code.R` takes the median of them on each axis:

```
direct.maxpvalue.2 <- median(directs[which(pmat.ssr == max(pmat.ssr), arr.ind = TRUE)[,1]])
```

`CoppockJEPS_figure3code.R` writes it differently, and so do the four appendix figure scripts and `CoppockJEPS_figure1bcode.R`:

```
low.direct.maxpvalue   <- directs[which(low.pmat.ssr == max(low.pmat.ssr), arr.ind = TRUE)[1]
low.indirect.maxpvalue <- indirects[which(low.pmat.ssr == max(low.pmat.ssr), arr.ind = TRUE)
```

`which(..., arr.ind = TRUE)` returns a two-column matrix of row and column indices, one row per tied cell. A single bracket reads it in column-major order, so `[1]` is the row index of the first tied cell and `[2]` is the row index of the *second* one, which the line above then uses to index the indirect grid. Where only one cell attains the maximum the two forms agree, which is why Figures 1b, A2 and A3 are unaffected. Figure 3's lower-support panel has 77 tied cells.

Table 3: Figure 3's point estimates, as the archive computes them and as the paper describes them.

Quantity	Published	Median of the tied cells
direct effect, lower support	-0.550	-0.4750
indirect effect, lower support	-0.125	-0.1750
direct effect, higher support	-0.175	-0.1875
indirect effect, higher support	-0.300	-0.3000

The confidence regions are unaffected, since they come from a different calculation, and the article's substantive claims about Figure 3 (the direct effect is bounded away from zero among lower-support districts and not among higher-support ones) hold under either reading.

5.2 The seed that is not set

`CoppockJEPS_appendixfigure4.R` never calls `set.seed`, alone among the archive's figure scripts. Appendix Figure 4 is therefore a different picture on every run, and the rewrite's version of it differs from the published one for that reason rather than any other. The rewrite seeds at 343, matching the rest of the archive. Nothing substantive turns on it: the geographic contiguity model is a placebo, and its confidence region contains zero on every run.

5.3 The simulation that never permutes

`CoppockJEPS_appendixfigure5.R` estimates the four-parameter generalization of Figure 3, in which the direct and indirect effects in higher- and lower-support districts are estimated jointly. Its simulation loop reads:

```
for(i in 1:sims){
  Z.sim <- Z_block[,1]
```

The assignment is column 1 on every iteration, so all fifty simulations of every one of the 28,561 cells use the same treatment vector, and every p-value in the array is either 0 or 1. The published Appendix Figure 5 shows this plainly: its shaded regions are solid black rectangles with no gradation at all, unlike every other p-value map in the paper. Every other script in the archive writes `Z_block[, sample(1:10000, 1)]` on that line.

The defect is a wrong index rather than an analytical choice, so the rewrite corrects it and computes both versions. The uncorrected slice takes 2 distinct values; the corrected one takes 20. The appendix states no numbers for this figure and Appendix Tables 1 and 2 do not cover it, so there is no published value the correction can be said to change; what it changes is that the figure now carries information. The appendix's claim that the four-parameter results are substantively the same as Figure 3's survives the correction.

6 Number-by-number comparison

Table 4: Ground truth: published value against the value the deposited scripts produce on current R. A blank match means the archive prints no comparable value.

Source	Location	Quantity	Published	Archive	Match
coppock_2014	figure_1a	direct effect	-0.16	-0.16	1
coppock_2014	figure_1a	direct 95% CI lower	-0.34	-0.34	1
coppock_2014	figure_1a	direct 95% CI upper	0.02	0.02	1
coppock_2014	figure_1b	effect, higher support	0.05	0.05	1
coppock_2014	figure_1b	higher support 95% CI lower	-0.225	-0.225	1
coppock_2014	figure_1b	higher support 95% CI upper	0.35	0.35	1
coppock_2014	figure_1b	effect, lower support	-0.375	-0.375	1
coppock_2014	figure_1b	lower support 95% CI lower	-0.675	-0.675	1
coppock_2014	figure_1b	lower support 95% CI upper	-0.05	-0.05	1
coppock_2014	figure_2	direct effect	-0.3	-0.3	1
coppock_2014	figure_2	direct 95% CI lower	-0.5	-0.5	1
coppock_2014	figure_2	direct 95% CI upper	-0.075	-0.075	1
coppock_2014	figure_2	indirect effect	-0.225	-0.225	1
coppock_2014	figure_2	indirect 95% CI lower	-0.3	-0.325	0
coppock_2014	figure_2	indirect 95% CI upper	-0.1	-0.1	1
coppock_2014	figure_3a	direct effect, lower support	-0.55	-0.55	1
coppock_2014	figure_3a	direct 95% CI lower	-0.825	-0.8	0
coppock_2014	figure_3a	direct 95% CI upper	-0.125	-0.125	1
coppock_2014	figure_3a	indirect effect, lower support	-0.125	-0.125	1
coppock_2014	figure_3a	indirect 95% CI lower	-0.325	-0.325	1
coppock_2014	figure_3a	indirect 95% CI upper	0	0	1
coppock_2014	figure_3b	direct effect, higher support	-0.175	-0.175	1
coppock_2014	figure_3b	direct 95% CI lower	-0.45	-0.4	0
coppock_2014	figure_3b	direct 95% CI upper	0.075	0.075	1
coppock_2014	figure_3b	indirect effect, higher support	-0.3	-0.3	1
coppock_2014	figure_3b	indirect 95% CI lower	-0.425	-0.425	1
coppock_2014	figure_3b	indirect 95% CI upper	-0.175	-0.175	1
coppock_2014	figure_a2	direct effect	-0.225	-0.225	1
coppock_2014	figure_a2	direct 95% CI lower	-0.45	-0.45	1
coppock_2014	figure_a2	direct 95% CI upper	0.1	0.1	1
coppock_2014	figure_a2	indirect effect	-0.2	-0.2	1
coppock_2014	figure_a2	indirect 95% CI lower	-0.325	-0.325	1
coppock_2014	figure_a2	indirect 95% CI upper	0.175	0.15	0
coppock_2014	figure_a3	direct effect	-0.25	-0.25	1
coppock_2014	figure_a3	direct 95% CI lower	-0.525	-0.525	1
coppock_2014	figure_a3	direct 95% CI upper	0.325	0.325	1
coppock_2014	figure_a3	indirect effect	-0.175	-0.175	1
coppock_2014	figure_a3	indirect 95% CI lower	-0.35	-0.325	0
coppock_2014	figure_a3	indirect 95% CI upper	0.525	0.525	1
coppock_2014	figure_a4	direct effect	-0.15	-0.15	1
coppock_2014	figure_a4	direct 95% CI lower	-0.38	-0.38	1
coppock_2014	figure_a4	direct 95% CI upper	0.08	0.08	1
coppock_2014	figure_a4	indirect effect	-0.025	-0.025	1
coppock_2014	figure_a4	indirect 95% CI lower	-0.095	-0.095	1
coppock_2014	figure_a4	indirect 95% CI upper	0.145	0.145	1
coppock_2014	text	legislators treated	35	35	1
coppock_2014	text	implied change in votes, difference in means	-5.6	-5.6	1
coppock_2014	text	yea votes in the uniformity trial	61		
coppock_2014	text	legislators voting yea	51		
coppock_2014	text	total causal effect in votes	10		
coppock_2016	figure_2 uncorrected	direct effect	-0.3	-0.3	1
coppock_2016	figure_2 uncorrected	direct 95% CI lower	-0.575	-0.5	0
coppock_2016	figure_2 uncorrected	direct 95% CI upper	0	-0.075	0
coppock_2016	figure_2 uncorrected	indirect effect	-0.225	-0.225	1
coppock_2016	figure_2 uncorrected	indirect 95% CI lower	-0.325	-0.325	1

Table 4: Ground truth: published value against the value the deposited scripts produce on current R. A blank match means the archive prints no comparable value. *(continued)*

Source	Location	Quantity	Published	Archive	Match
coppock_2016	figure_2 uncorrected	indirect 95% CI upper	-0.025	-0.1	0
coppock_2016	figure_2 corrected	direct effect	-0.25		
coppock_2016	figure_2 corrected	direct 95% CI lower	-0.625		
coppock_2016	figure_2 corrected	direct 95% CI upper	0.175		
coppock_2016	figure_2 corrected	indirect effect	-0.15		
coppock_2016	figure_2 corrected	indirect 95% CI lower	-0.325		
coppock_2016	figure_2 corrected	indirect 95% CI upper	0.2		

Of the 62 recorded claims, 53 can be compared against what a deposited script prints, and 45 of those match the published value. Every failure is either a confidence region endpoint moved one grid step by the sampler change, or one of the corrigendum's restated intervals discussed above.

7 Maintained rewrite

The rewrite lives in `maintained/`: one cleaning script, nine figure scripts, two table scripts and one script for the in-text quantities.

7.1 Architecture

`clean_data.R` is the load-bearing one. It reads the deposited `CoppockJEPS.RData`, rebuilds all four similarity matrices from the deposited ideal points and shapefile, and prints how far each rebuilt matrix sits from the deposited one, which is what localizes the defect: three deviations of zero and one of 1. It then computes expected and observed net exposures for each matrix, and keeps the deposited all-ones matrix alongside the corrected one so that the uncorrected figures can be reproduced rather than described.

Every spillover figure is therefore computed twice, once from each matrix, and both versions go into `output/`. Nothing is transcribed: the corrigendum's numbers appear in the ground truth as comparison targets and nowhere in the code.

The p-value grid itself is one function, `pvalue_grid()` in `helpers.R`, taking the similarity matrix, the expected exposures and the observed exposure as separate arguments. That separation is deliberate. The difference between the published analysis and the corrected one is exactly which matrix goes into the simulation while the observed regressor stays the same, and a helper that fixed either of them internally would hide the only choice in the paper that matters.

The rewrite's p-value matrices were checked cell by cell against the ones the deposited scripts save. Six of the seven are bit-identical: Figure 1a's 71 cells, Figure 1b's 2,205, Figure 2's 1,369,

Figure 3's 1,305, Appendix Figure 2's 1,681 and Appendix Figure 3's 2,401, with a maximum absolute deviation of exactly zero in every case. Appendix Figure 4 is the exception, and only because its script sets no seed. Agreement at that level is what makes the point-estimate differences interpretable: the surfaces are the same, so any difference in what is read off them is a difference in the reading.

The test statistic goes through `.lm.fit` rather than `lm`. The rewrite fits about eleven million regressions; `lm` takes 0.295 milliseconds each and `.lm.fit` takes 0.0035, and the two agree on the sum of squared residuals to floating point. That is the difference between a five-minute run and an hour-long one, and a reproduction check nobody runs is not a check.

7.2 Deprecated patterns replaced

Table 5: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd(”“)‘ and ‘Replication Archive/‘ paths	‘here::here()‘
‘maptools::readShapeSpatial‘	‘sf::st_read‘ + ‘spdep::poly2nb‘
‘lattice::levelplot‘ + ‘larrows‘	‘ggplot2::geom_tile‘ + ‘annotate(”segment”)‘
‘pdf()‘ / ‘dev.off()‘	‘ggsave()‘, PDF and PNG
‘xtable‘ + ‘sink()‘ to a hardcoded path	‘write_csv()‘ to ‘output/‘
‘save()‘ / ‘load()‘ of .rdata	‘write_rds()‘ / ‘read_rds()‘
nested ‘for‘ loops over ‘data.frame(...)‘ + ‘lm()‘	design matrices + ‘.lm.fit‘
‘Z_block[, 1]‘ inside the simulation loop	‘z_block[, sample(ncol(z_block), 1)]‘

7.3 Summary tables

Table 6: Appendix Table 1, constant effects models, as the maintained rewrite produces it.

	No spillover	Absolute distance	Absolute distance (uncorrected)	Rank distance	Squared distance	Geographic
Direct Effect	-0.16	-0.2875	-0.3	-0.225	-0.25	-0.15
Direct 95% CI	[-0.34,0.02]	[-0.5,0.075]	[-0.5,-0.075]	[-0.45,0.1]	[-0.525,0.325]	[-0.37,0.09]
Indirect Effect		-0.225	-0.225	-0.2	-0.175	0.025
Indirect 95% CI		[-0.3,0.15]	[-0.325,-0.1]	[-0.325,0.15]	[-0.325,0.525]	[-0.095,0.15]
Corresponding Figure	1a	2	2	A2	A3	A4

Table 7: Appendix Table 2, heterogeneous effects models, as the maintained rewrite produces it.

	No spillover	Absolute distance	Absolute distance (uncorrected)
Direct Effect, High Support	0.05	-0.2	-0.1875
Direct 95% CI, High Support	[-0.225,0.35]	[-0.45,NA]	[-0.4,0.075]
Indirect Effect, High Support		-0.3	-0.3
Indirect 95% CI, High Support		[-0.425,0.175]	[-0.425,-0.175]

Direct Effect, Low Support	-0.375	-0.45	-0.475
Direct 95% CI, Low Support	[-0.675,-0.05]	[-0.75,-0.05]	[-0.8,-0.125]
Indirect Effect, Low Support		-0.15	-0.175
Indirect 95% CI, Low Support		[-0.325,0.125]	[-0.325,0]
Corresponding Figure	1b	3	3

7.4 In-text quantities

Table 8: The page 167 arithmetic, under both similarity matrices.

Quantity	Value
Legislators treated	35
Difference-in-means estimate (Figure 1a)	-0.16
Implied change in votes under the difference in means	-5.6
Legislators voting yea	51
Yea votes in the uniformity trial, uncorrected	61.46
Total causal effect in votes, uncorrected	10.46
Yea votes in the uniformity trial, corrected	61.02
Total causal effect in votes, corrected	10.02

The article states that 61 legislators would have voted yea in the uniformity trial against 51 who actually did, for a total causal effect of 10 votes, roughly twice the 5.6 the difference in means implies. The uncorrected column reproduces that. The corrected column is the quantity the corrigendum's conclusion bears on: the point estimates barely move, so the arithmetic barely moves, and what changes is that neither effect can be distinguished from zero.

8 Figure verification

The correction's visible consequence is the width of the shaded region. The uncorrected map is a tight blob well away from both axes; the corrected map spreads across zero on both.

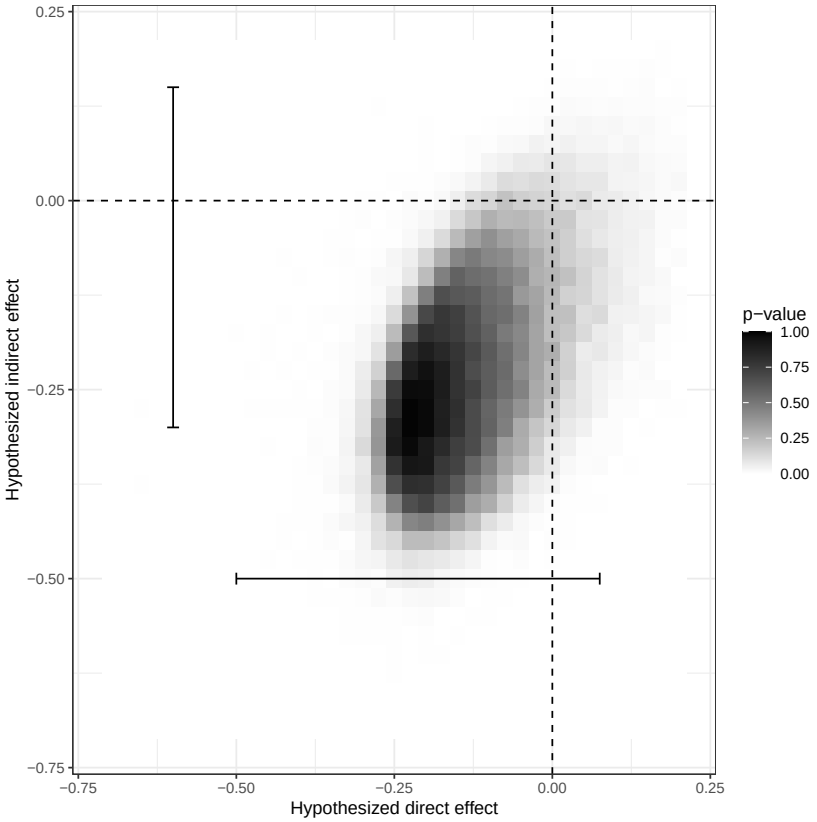


Figure 1: Figure 2, corrected similarity matrix.

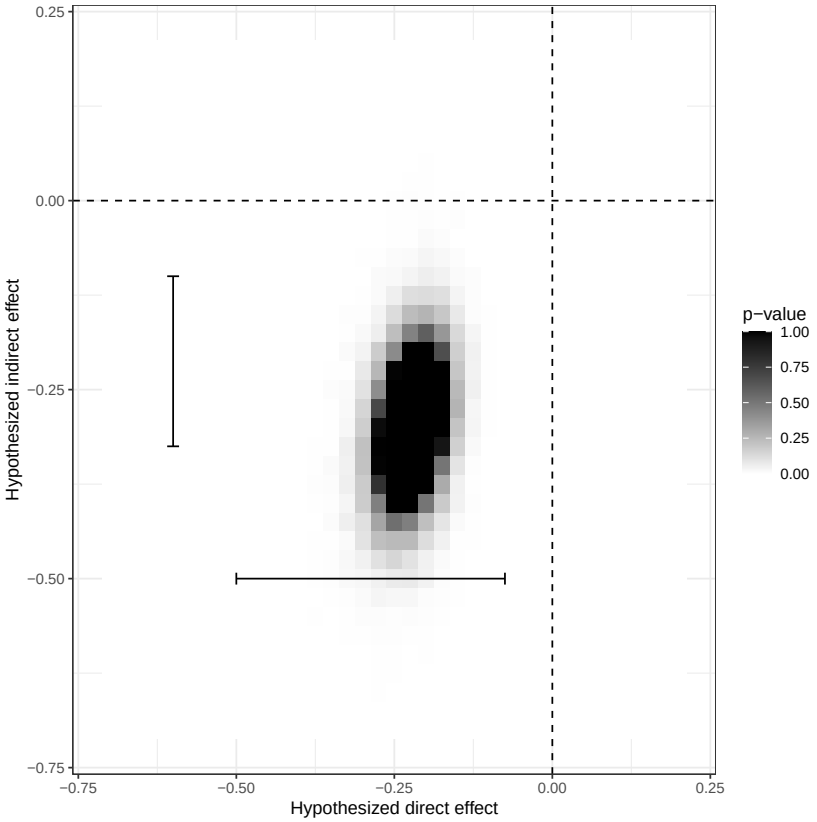


Figure 2: Figure 2 as published, all-ones similarity matrix.

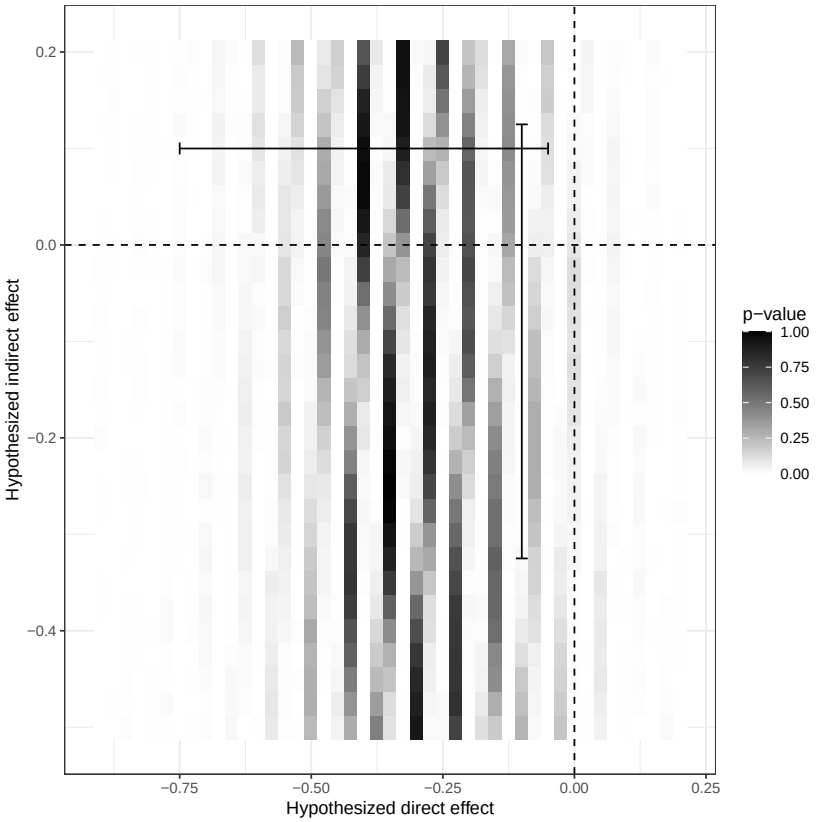


Figure 3: Figure 3a, lower-support districts, corrected.

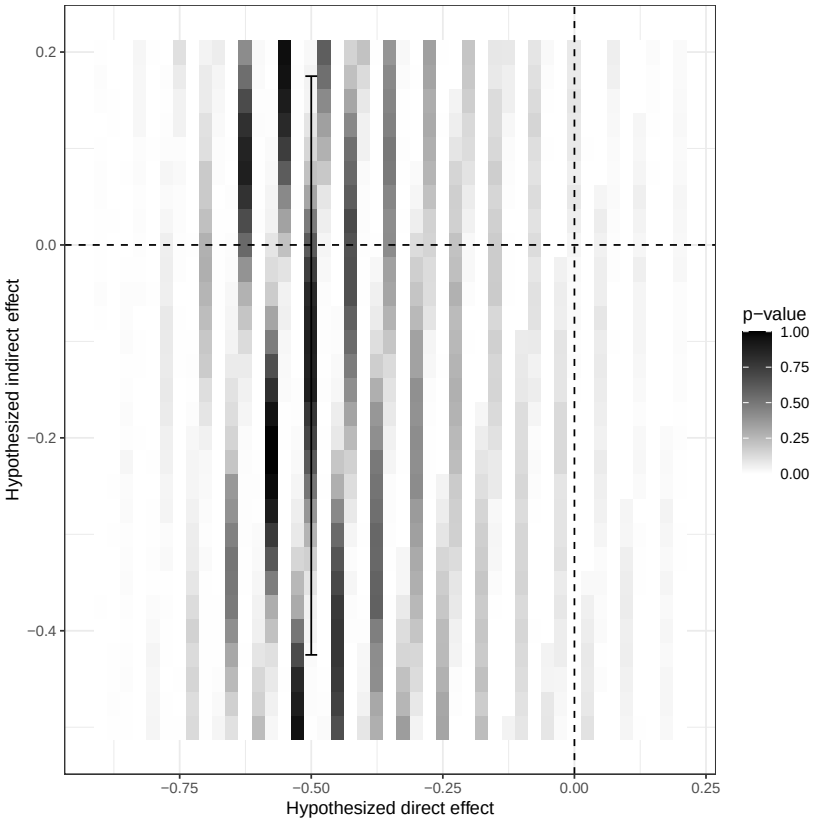


Figure 4: Figure 3b, higher-support districts, corrected.

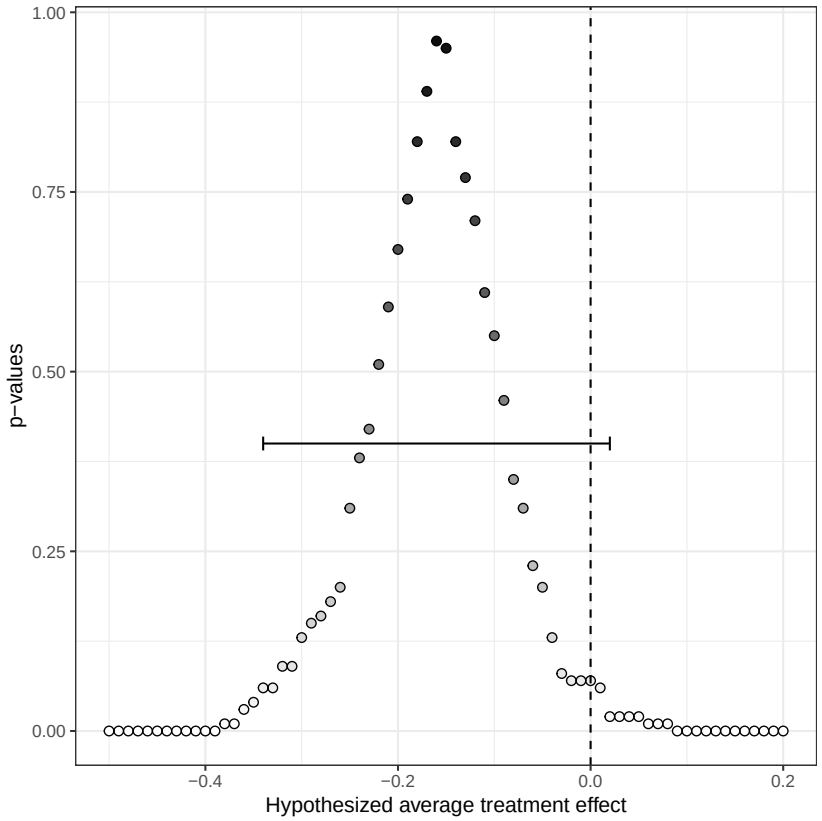


Figure 5: Figure 1a, the no-spillover constant effects model, which reproduces exactly.

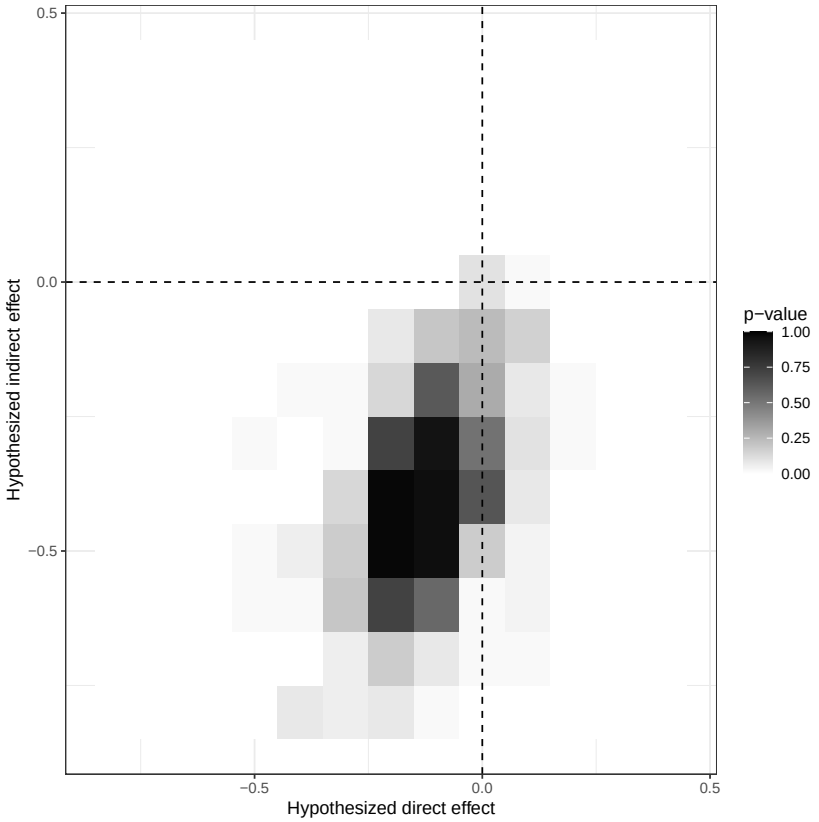


Figure 6: Appendix Figure 5a with a fresh assignment drawn per simulation.

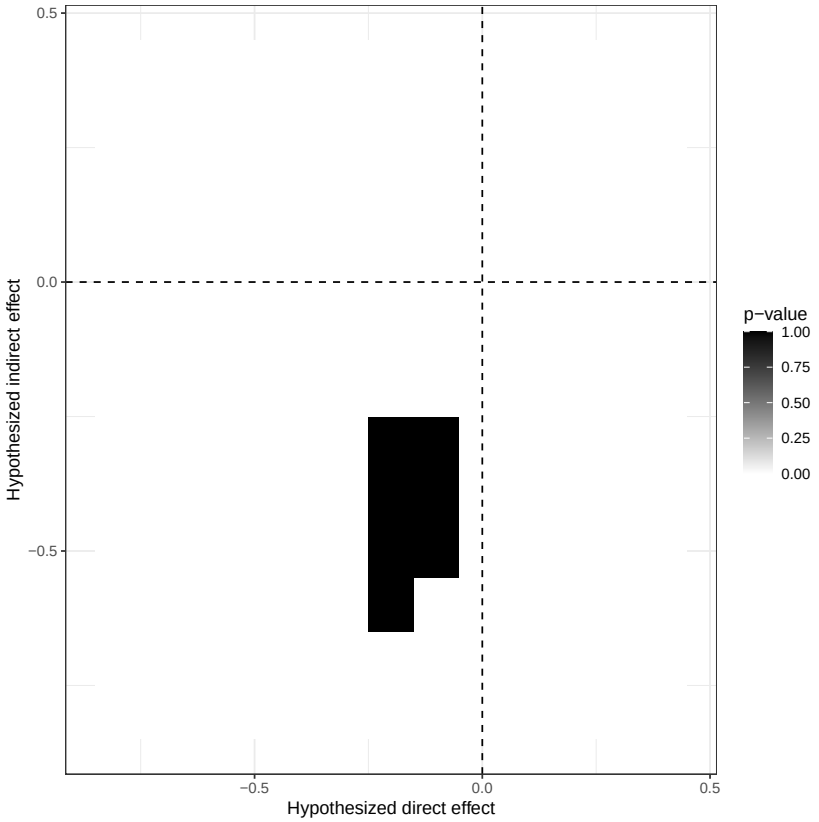


Figure 7: Appendix Figure 5a as published: every p-value is 0 or 1.

9 Maintained rewrite verification

Table 9: Maintained rewrite verification: published value against rewrite output.

Source	Location	Quantity	Published	Rewrite	Match
coppock_2014	figure_1a	direct effect	-0.16	-0.16	1
coppock_2014	figure_1a	direct 95% CI lower	-0.34	-0.34	1
coppock_2014	figure_1a	direct 95% CI upper	0.02	0.02	1
coppock_2014	figure_1b	effect, higher support	0.05	0.05	1
coppock_2014	figure_1b	higher support 95% CI lower	-0.225	-0.225	1
coppock_2014	figure_1b	higher support 95% CI upper	0.35	0.35	1
coppock_2014	figure_1b	effect, lower support	-0.375	-0.375	1
coppock_2014	figure_1b	lower support 95% CI lower	-0.675	-0.675	1
coppock_2014	figure_1b	lower support 95% CI upper	-0.05	-0.05	1
coppock_2014	figure_2	direct effect	-0.3	-0.3	1
coppock_2014	figure_2	direct 95% CI lower	-0.5	-0.5	1
coppock_2014	figure_2	direct 95% CI upper	-0.075	-0.075	1
coppock_2014	figure_2	indirect effect	-0.225	-0.225	1
coppock_2014	figure_2	indirect 95% CI lower	-0.3	-0.325	0
coppock_2014	figure_2	indirect 95% CI upper	-0.1	-0.1	1
coppock_2014	figure_3a	direct effect, lower support	-0.55	-0.475	0

Table 9: Maintained rewrite verification: published value against rewrite output. *(continued)*

Source	Location	Quantity	Published	Rewrite	Match
coppock_2014	figure_3a	direct 95% CI lower	-0.825	-0.8	0
coppock_2014	figure_3a	direct 95% CI upper	-0.125	-0.125	1
coppock_2014	figure_3a	indirect effect, lower support	-0.125	-0.175	0
coppock_2014	figure_3a	indirect 95% CI lower	-0.325	-0.325	1
coppock_2014	figure_3a	indirect 95% CI upper	0	0	1
coppock_2014	figure_3b	direct effect, higher support	-0.175	-0.1875	0
coppock_2014	figure_3b	direct 95% CI lower	-0.45	-0.4	0
coppock_2014	figure_3b	direct 95% CI upper	0.075	0.075	1
coppock_2014	figure_3b	indirect effect, higher support	-0.3	-0.3	1
coppock_2014	figure_3b	indirect 95% CI lower	-0.425	-0.425	1
coppock_2014	figure_3b	indirect 95% CI upper	-0.175	-0.175	1
coppock_2014	figure_a2	direct effect	-0.225	-0.225	1
coppock_2014	figure_a2	direct 95% CI lower	-0.45	-0.45	1
coppock_2014	figure_a2	direct 95% CI upper	0.1	0.1	1
coppock_2014	figure_a2	indirect effect	-0.2	-0.2	1
coppock_2014	figure_a2	indirect 95% CI lower	-0.325	-0.325	1
coppock_2014	figure_a2	indirect 95% CI upper	0.175	0.15	0
coppock_2014	figure_a3	direct effect	-0.25	-0.25	1
coppock_2014	figure_a3	direct 95% CI lower	-0.525	-0.525	1
coppock_2014	figure_a3	direct 95% CI upper	0.325	0.325	1
coppock_2014	figure_a3	indirect effect	-0.175	-0.175	1
coppock_2014	figure_a3	indirect 95% CI lower	-0.35	-0.325	0
coppock_2014	figure_a3	indirect 95% CI upper	0.525	0.525	1
coppock_2014	figure_a4	direct effect	-0.15	-0.15	1
coppock_2014	figure_a4	direct 95% CI lower	-0.38	-0.37	0
coppock_2014	figure_a4	direct 95% CI upper	0.08	0.09	0
coppock_2014	figure_a4	indirect effect	-0.025	0.025	0
coppock_2014	figure_a4	indirect 95% CI lower	-0.095	-0.095	1
coppock_2014	figure_a4	indirect 95% CI upper	0.145	0.15	0
coppock_2014	text	legislators treated	35	35	1
coppock_2014	text	implied change in votes, difference in means	-5.6	-5.6	1
coppock_2014	text	yea votes in the uniformity trial	61	61.46	0
coppock_2014	text	legislators voting yea	51	51	1
coppock_2014	text	total causal effect in votes	10	10.46	0
coppock_2016	figure_2 uncorrected	direct effect	-0.3	-0.3	1
coppock_2016	figure_2 uncorrected	direct 95% CI lower	-0.575	-0.5	0
coppock_2016	figure_2 uncorrected	direct 95% CI upper	0	-0.075	0
coppock_2016	figure_2 uncorrected	indirect effect	-0.225	-0.225	1
coppock_2016	figure_2 uncorrected	indirect 95% CI lower	-0.325	-0.325	1
coppock_2016	figure_2 uncorrected	indirect 95% CI upper	-0.025	-0.1	0
coppock_2016	figure_2 corrected	direct effect	-0.25	-0.2875	0
coppock_2016	figure_2 corrected	direct 95% CI lower	-0.625	-0.5	0
coppock_2016	figure_2 corrected	direct 95% CI upper	0.175	0.075	0
coppock_2016	figure_2 corrected	indirect effect	-0.15	-0.225	0
coppock_2016	figure_2 corrected	indirect 95% CI lower	-0.325	-0.3	0
coppock_2016	figure_2 corrected	indirect 95% CI upper	0.2	0.15	0

39 of the 62 verifiable claims match to reported precision. Every mismatch carries a note naming its cause, and none of them is a defect in the rewrite: confidence region endpoints moved one grid step by the R 3.6 sampler; the four Figure 3 point estimates and the whole of Appendix Figure 4, both covered under “Errata”; the corrigendum’s restated intervals for the uncorrected model; and the corrigendum’s corrected values, which no rerun of the deposited data recovers.

10 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-01

Table 11: Package versions used for the run behind this report.

Package	Version
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
purrr	1.2.2
readr	2.2.0
sf	1.1.2
spdep	1.4.2
here	1.0.2